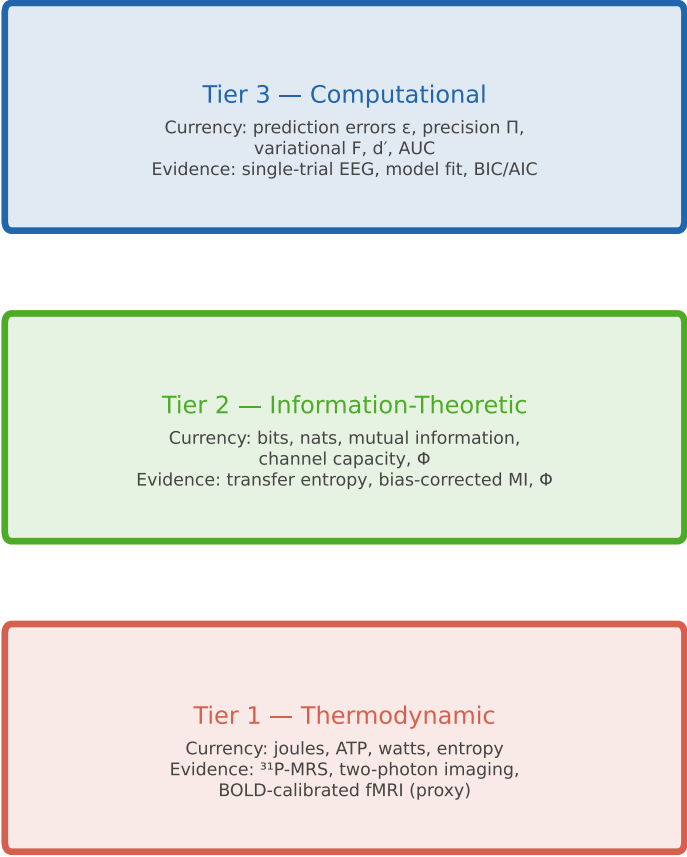


Three-Tier Epistemic Architecture

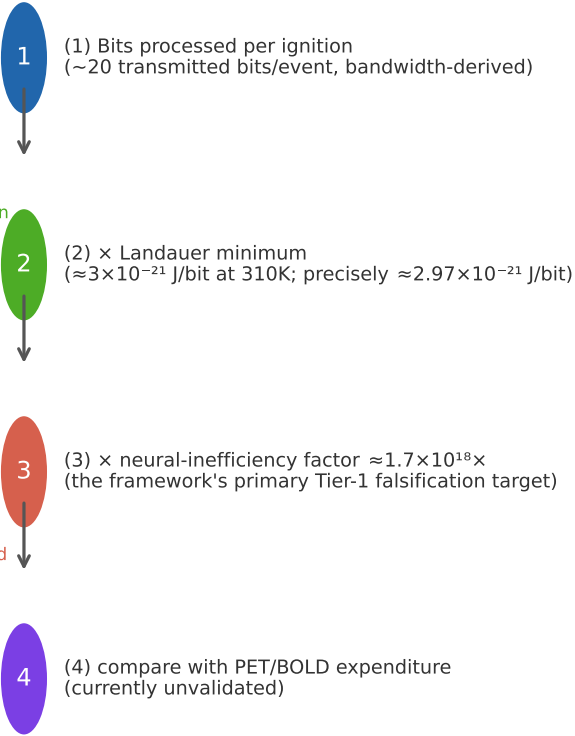


Valid bridges (solid)
Absent bridges (dashed $\times$)

T2→T3
Not yet specified $\times$

T1→T2
Not yet specified $\times$

Double-bridge calculation



T3→T2: Approximate-Inference Bridge
variational F minimisation $\rightarrow$ MI maximisation
Status: derived in ML, unconverged in vivo

T2→T1: Landauer bridge
 $E \geq kT \cdot \ln 2$ per bit erased
 $k \approx 100$ ATP/bit $\approx 5 \times 10^{-18}$ J/bit
Status: theoretically bounded, k unmeasured

T3→T1: Double bridge (3→2→1)
requires both above
Status: composite, unvalidated

Footnote: phenomenal level L4 (Φ -boundary) is excluded by design (§8.4)